Exercise 6.8: Do the same for the other two entangled triplet states,

$$|T_2\rangle = \frac{1}{\sqrt{2}}\Big(|uu\rangle + |dd\rangle\Big)$$

$$|T_3\rangle = \frac{1}{\sqrt{2}}\Big(|uu\rangle - |dd\rangle\Big),$$

and interpret.

Finally, let's consider one more observable. This one cannot be measured by Alice and Bob making separate measurements with their individual apparatuses, even if they come together and compare notes. Nevertheless, quantum mechanics insists that some kind of apparatus can be built to measure the observable.

The observable I am referring to can be thought of as the ordinary dot product of the vector-operators $\vec{\sigma}$ and $\vec{\tau}$:

$$\vec{\sigma} \cdot \vec{\tau} = \sigma_x \tau_x + \sigma_y \tau_y + \sigma_z \tau_z .$$

One might think that a value for this observable can be found if Bob measures all components of τ, while Alice measures all components of σ; then they could multiply the components and add them up. The problem is that Bob cannot simultaneously measure the individual components of τ, because they don't commute. Likewise, Alice cannot measure more than one component of σ at a time. To measure $\vec{\sigma} \cdot \vec{\tau}$, a new kind of apparatus must be built, one that measures $\vec{\sigma} \cdot \vec{\tau}$ without measuring any individual component. It's far from obvious how that could be done. Here is a concrete example of how such a measurement could be carried out: Some atoms have spins that are described in the same way as electron spins. When two of these atoms are close to each other—for example, two neighboring atoms in a crystal lattice—the Hamiltonian will depend on the spins. In some